

# EngageIQ — Smart Engagement Opportunity Scorer

## Shivneel Prasad · BAX-423 Big Data · Spring 2026

Live URL: <https://engageiq-bax423-final.streamlit.app/>  
GitHub: <https://github.com/shprasa/engageiq-bax423-final>

### 1. System architecture & pipeline

Pipeline: GitHub API + Hacker News API ingestion (PRAW optional) → Bloom-filter deduplication → Count-Min Sketch + HyperLogLog streaming metrics → DuckDB batch store → TF-IDF/SVD embeddings → cosine ANN retrieval → multi-stage ranker (relevance, health, visibility, effort, recency) → reinforcement learning policy (Thompson-sampling contextual bandit) from engage/skip/bookmark feedback → Streamlit dashboard with RL policy view, trend analytics, and CSV/PDF brief export. Offline snapshot: 11964 rows (1964 live API + 10000 synthetic backup), 15 domains. Live sources: {'hackernews': 1036, 'github': 928}.

### 2. BAX-423 techniques & benchmarks

**Sketching:** Bloom dedup detected 18 duplicate URLs; CMS/HLL power dashboard trend metrics. **Embeddings + ranking:** TF-IDF/SVD + NearestNeighbors; NDCG@10 reported per persona. **Reinforcement learning:** contextual multi-armed bandit (Thompson sampling) — state=user profile+history, actions=domain recommendations, rewards from engage/bookmark/skip. 60-round benchmark: NDCG@10 with RL 1.000 vs without 1.000; cumulative reward 60 vs 60; reward Δ(last 10)=+0.000.

### 3. Persona results (top-10 checks)

| Persona | NDCG@10 | Checks                 | Overall |
|---------|---------|------------------------|---------|
| Sofia   | 1.000   | GFI=0, C++/Rust=0      | PARTIAL |
| David   | 0.000   | Infra=9/10             | PASS    |
| Lina    | 0.000   | Visibility ≥ relevance | PASS    |
| Raj     | 1.000   | DevTools=10/10         | PASS    |

### 4. Persona × capability pass/fail (required)

| Persona | Multi-source ingest | Embeddings + ANN | Scoring + multi-stage | Adaptive learning / RL | Batch analytics + trend | Dashboard + brief expo |
|---------|---------------------|------------------|-----------------------|------------------------|-------------------------|------------------------|
| Sofia   | PASS                | PASS             | PARTIAL               | PARTIAL                | PASS                    | PASS                   |
| David   | PASS                | PASS             | PASS                  | PARTIAL                | PASS                    | PASS                   |
| Lina    | PASS                | PASS             | PASS                  | PARTIAL                | PASS                    | PASS                   |
| Raj     | PASS                | PASS             | PASS                  | PARTIAL                | PASS                    | PASS                   |

### 5. Limitations

Reddit live ingestion skipped (no PRAW creds); synthetic rows ensure 10k offline grading. Embeddings use TF-IDF/SVD vs Sentence-BERT/FAISS for deploy reliability. Suggested actions are template-based (LLM optional). GH Archive/Kafka included as optional streaming extensions.